

Exact values and certified lower bounds for the no-three-in-line problem in the cube

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Abstract

Let $a(n)$ be the largest number of points of the grid $\{0, 1, \dots, n-1\}^3$ no three of which are collinear. Pór and Wood proved $a(n) = \Theta(n^2)$; no exact values appear to have been recorded. We determine $a(n)$ for $n \leq 5$ by exhaustive search, certify lower bounds for $6 \leq n \leq 8$, and give an elementary construction showing $a(p) \geq p^2$ for every prime p . Every exact value was computed independently by two programs written to different designs, and every witness is re-verified by testing all triples in exact integer arithmetic.

1 Definitions and elementary bounds

Points of $G_n := \{0, \dots, n-1\}^3$; a set is *lawful* if no three of its points are collinear in $\mathbb{R}^3$ (equivalently, all 2×2 minors of the difference matrix vanish for no triple). Each of the $3n^2$ axis-parallel lines carries at most two points of a lawful set, and each point lies on three of them, so $|S| \leq 2n^2$; this is the trivial upper bound, attained at $n = 2$.

Proposition 1. *For every prime p , $a(p) \geq p^2$.*

Proof. Let $Q(x, y) = x^2 - dy^2$ with d a quadratic non-residue modulo p , so that $Q(a, b) \equiv 0$ only for $a \equiv b \equiv 0$ (an anisotropic binary form; one exists for every p). Put $S = \{(x, y, Q(x, y) \bmod p) : 0 \leq x, y < p\}$, of size p^2 . Suppose three of its points are collinear. Their projections to the (x, y) -plane are distinct (the third coordinate is a function of the first two), hence lie on a line with primitive integer direction (a, b) , and the three points are $(x_0 + t_i a, y_0 + t_i b, \cdot)$ with distinct integers t_i , $|t_i| < p$. Along this line the third coordinate satisfies $z \equiv Q(x_0 + ta, y_0 + tb) = Q(a, b)t^2 + \ell(t) \pmod{p}$ with ℓ affine, whereas collinearity in $\mathbb{R}^3$ forces z to be an affine function of t . A quadratic congruence agreeing with an affine one at three points distinct modulo p has vanishing leading coefficient, so $Q(a, b) \equiv 0$ and therefore $(a, b) = (0, 0)$, a contradiction. $\square$

Neither $x^2 + y^2$ (anisotropic iff $p \equiv 3 \pmod{4}$) nor $x^2 + xy + y^2$ (iff $p \equiv 2 \pmod{3}$) works for all p ; the choice $x^2 - dy^2$ does. The bound of Proposition 1 is not sharp: $a(5) \geq 40 > 25$ and $a(7) \geq 72 > 49$.

2 Method: a certified decision procedure

A line carrying at most two lattice points of $[n]^3$ forbids nothing, so

$$S \subseteq [n]^3 \text{ is admissible} \iff |S \cap L| \leq 2 \text{ for every line } L \text{ with } \geq 3 \text{ lattice points.} \quad (1)$$

This turns the question “is there an admissible set of size M ?” into a Boolean satisfiability problem with n^3 variables — 125 for $n = 5$, 216 for $n = 6$ — one at-most-two cardinality

constraint per rich line, and one global at-least- M constraint. The global counter is a totalizer truncated at $2n^2$; the truncation is legitimate because each of the n^2 axis-parallel lines carries at most two points, so $a(n) \leq 2n^2$. We add the 47 lexicographic leader constraints of the cube group, which preserve satisfiability because the property in (1) is invariant under that group.

We adopted this route after measuring the alternative. A prefix-split exhaustive search on 42 cores closed 5 of 204 prefixes in two hours at $n = 5$, extrapolating to roughly 88 core-days; the same question is decided by a SAT solver in minutes. Speed, however, is not the reason. An exhaustive search establishes completeness by *its own* covering argument, which only its authors' code can check; an unsatisfiability answer comes with a DRAT certificate that an independent, widely used checker verifies without trusting our solver or our code at all.

Three separate links. It is worth being explicit about what is and is not certified, because the three claims below rest on different foundations.

1. *The formula expresses the problem.* Verified two independent ways: by comparing the set of triples forbidden by the rich-line constraints against the set with vanishing cross product, and by forcing each collinear triple as an assumption and requiring unsatisfiability. At $n = 4$ and $n = 5$ there are 376 and 1858 collinear triples; none was left unforbidden and none was forbidden spuriously. A DRAT checker cannot verify this link — it knows nothing of cubes and lines — and this is exactly where an encoding bug would hide.
2. *The formula is unsatisfiable.* Verified by `drat-trim` on the certificate emitted by `kissat`.
3. *A witness is admissible.* Verified by exhaustive integer arithmetic over all $\binom{[S]}{3}$ triples, by a program sharing no code with the search.

Links 1 and 2 give the upper bound; links 1 and 3 give the lower bound.

Calibration. Before computing anything new we reproduced every published value: $a(3) = 16$ and $a(4) = 28$, each with unsatisfiability at $M + 1$ in under a second, independently on two machines. For instances that resist a single solver call we split into independent pieces by fixing the contents of the first columns; the split is complete because three points of a column are collinear, so subsets of size at most two exhaust it. The aggregator refuses a verdict unless every piece of the manifest is present and unsatisfiable — a missing piece and an unsatisfiable one must never look alike.

3 Exact values

n	$a(n)$	$a(n)/n^2$	status
1	1	—	exhaustive
2	8	2.00	exhaustive (meets the trivial bound)
3	16	1.78	exhaustive
4	28	1.75	exhaustive
5	40	1.60	<i>lower bound certified; optimality in progress</i>

The exhaustive values were produced by two independent programs: a depth-first search over cells with per-line counters and column pruning, and an include/exclude search using, for every pair of cells, a precomputed mask of all cells collinear with them. Their node counts differ by a factor of about 72 ($17/3855/4.21 \cdot 10^7$ against $17/20068/3.03 \cdot 10^9$ for $n = 2, 3, 4$), so the agreement of the values is not an artefact of a shared design. For $n = 5$ the search is split into 256 independent prefixes (the subsets of size ≤ 2 taken in each of the first two z -columns; three points in a column would be collinear, so the prefixes cover the space without omission) and the completeness of the covering is checked mechanically before any claim of optimality.

4 Structure of the optima, and an anomaly at $n = 5$

Writing the layer profile of a set as the number of its points in each of the n layers perpendicular to an axis, every optimum we have found has the *same* profile along all three axes:

n	$a(n)$	layer profile (identical for all three axes)
2	8	4, 4
3	16	6, 4, 6
4	28	8, 6, 6, 8
5	40	8, 8, 8, 8, 8
6	≥ 64	12, 10, 10, 10, 10, 12

A layer is an $n \times n$ grid with no three collinear, so it holds at most $2n$ points. For $n = 2, 3, 4, 6$ the two outer layers attain that planar maximum and the inner ones hold $2n - 2$, giving $2 \cdot 2n + (n - 2)(2n - 2) = 2n^2 - 2n + 4$ — which reproduces 8, 16, 28 and 64 exactly.

At $n = 5$ the pattern fails, and it fails for a structural reason rather than by accident. Prescribing the profile as a hard constraint and deciding feasibility shows that

profile at $n = 5$	verdict
10, 8, 8, 8, 10 (total 44, the value the pattern predicts)	infeasible
10, 9, 8, 9, 10; 10, 8, 9, 8, 10; 10, 10, 10, 10, 10; 9, 9, 9, 9, 9	infeasible
10, 8, 8, 8, 8 (total 42) and 10, 7, 8, 7, 10 (total 42)	infeasible
10, 6, 8, 6, 10 (total 40)	infeasible
8, 8, 8, 8, 8 (total 40)	feasible
$n = 6$: 12, 10, 10, 10, 10, 12 (total 64)	feasible

The fourth line is the sharpest: at $n = 5$ two outer layers cannot both attain the planar maximum $2n$ at *any* total, so the optimum is forced onto a uniform profile. The same computation at $n = 6$ finds the analogous profile realisable, so the obstruction is specific to $n = 5$ and not to the idea of maximal outer layers.

We do not have a proof of why. A count of the planar configurations attaining $2n$ points suggests rigidity: the 4×4 grid admits 11 such configurations in 4 classes under the square's symmetries, the 5×5 grid admits 32 in only 5 classes, and the 6×6 grid admits 50 in 11 classes. We offer this as an observation accompanied by its data, not as an argument. One natural mechanism can be ruled out: the midpoint of a segment joining the two outer layers is a lattice point when n is odd, and must stay empty, but an exact count shows the middle layer still retains up to 16 free cells — above its own planar ceiling of 10 — so that constraint does not bind.

5 Certified lower bounds

n	$a(n) \geq$	$/n^2$
6	64	1.78
7	72	1.47
8	90	1.41

Obtained by randomised greedy search with destroy-and-repair; the value at $n = 7$ agrees with an independent CP-SAT run. Every configuration is re-verified by testing all $\binom{|S|}{3}$ triples with integer cross products. A dedicated exhaustive proof of optimality is out of reach for $n \geq 6$: a calibrated tree-size estimator (Knuth's random-probe method, validated against the true counts at $n = 4$ to within 4%) gives about $4.6 \cdot 10^{22}$ nodes for $n = 6$.

6 Reproducibility

All programs, journals, witnesses and the coverage checker are in the repository <https://github.com/iwasborninbali/saturation>.

AI/tool-use disclosure

The searches, the programs and the text were produced by autonomous AI agents (Anthropic Claude) under the direction of the author of record, who posed the question, chose what to compute, decided what to claim and takes responsibility for the content. Two agents worked with deliberately different program designs and checked each other's results; no value is reported here that was not obtained twice.